

CS 585: Computer Networks, Fall 2026

Project Proposal

Thursday, September 10, 2026

Team Members

- Natalie Runyan, Master's Program, Computer Science
- Carly Salazar, Master's Program, Computer Science
- Quinn Sena, PhD Program, Computer Science
- Joel Villarreal, Master's Program, Computer Science

1 Proposed Topic and Motivation

1.1 Topic

Meshcore is an open source protocol of communication that implements an ad hoc network system that can transmit messages to other users in range through radio waves. Meshcore allows a user to connect a transceiver node to a personal device to transmit or receive a message without access to the internet. Our group plans to investigate the security and efficacy in allowing users to communicate, search, and render information on Meshcore devices without the direct use of internet through an Internet Service Provider (ISP). Our group seeks to create a system where users of our mesh network are able to communicate to a central node wherein that node is able to parse a question a user sends and reach the external website without a user ever having to have direct contact with an ISP. Due to concerns of censorship or user not having a reliable connection to the internet, being able to search for information can allow users to access first aid information, messaging sites, news sites, and many more essential resources that are beyond valuable in an emergency. Our group also aims to provide benchmarking statistics as well as test the security of the Mesh Core nodes and repeaters to packet sniffing, packet overloading, and other security vulnerability's associated with radio communication. The benchmarking will show the efficacy of communication over mesh radio in New Mexico's unique terrain, as well as the most efficient place to place a mesh repeater.

1.2 Motivation

The internet has become an essential part of our daily lives as it facilitates communication and access to information. One cannot access this web of information without an intermediary Internet Service Provider. ISPs hold a strong hand in the modern information landscape, and their ability to track and sell user data should not be taken lightly. The right to privacy is something which large tech companies and ISPs are consistently attempting to undermine. For example, when an ISP is involved in Internet connection (all the time), an individual cannot access the internet without a user profile being built. To address this, our group aims to provide a method of private communication and web browsing via packeted radio connections. We hope that this will provide (1) security, as the connection can only be traced back to a willing gateway device (one connected to internet), and (2) be robust to failure, as any device with an Internet connection can feasibly become a gateway device or stop being a gateway device at any time. Additionally, we would

be remiss if we did not mention that the MeshCore devices which we plan to utilize can and are already used as communication devices without any need for the Internet. This fact motivates our preliminary security/benchmark testing. Concisely, our goal is to both test the performance of an already existing means of communication non-reliant on Internet, and to see if such a tool can be used to bolster user privacy when surfing the web.

1.3 Other topics

A couple other topic ideas our team finds interesting that we could pivot to if our proposed topic does not work out include:

- Radio jamming in a mesh network
- New Mexico based environmental benchmarking

2 Budget Draft

This section includes a list of resources, their proposed uses in the project, a proposed budget, as well as alternatives if funding is not received.

2.1 Resource Needs and Uses

What follows is a overly-thorough list of resources and their proposed uses in our project.

- **Computers** (Mobile, Desktop, Laptop, etc.): these are our interface with the network, akin to end-devices in an Internet network. For this project, computers will serve several purposes: typing messages to send over the mesh network; writing code; accessing the Internet; etc.
- **MeshCore App**: an application which lives on our computers. For this project, the MeshCore App will serve several purposes: connect computers to nodes; a subject for penetration testing; etc.
- **MeshCore Nodes**: these are the access points of the network, akin to a personal router in an Internet network. For this project, nodes will serve several purposes: messaging; benchmarking tests; subject for penetration testing; etc.
- **MeshCore Repeaters**: these are means of extending/bolstering the network, akin to a public access router in an Internet network. For this project, repeaters serve several purposes: extending range of messaging; adding more dimensions of variability to benchmarking tests; a subject of penetration testing; etc.
- **The Internet**: for this project, the Internet will serve as a means of researching (using tools such as Google Scholar, ILLiad, Google, YouTube, etc.).
- **Overleaf**: for writing the final paper.
- **PowerPoint/Google Slides**: for creating the final presentation.
- **GitHub**: for this project GitHub will serve a few purposes: access to existing open source code; a place to publish newly developed code.
- **Transportation**: vehicles will be a means to quickly adjust the configuration of the network (i.e. physically move repeaters and nodes).

- **Office Space:** for this project, we will hold our in-person meetings in FEC 3135 or other convenient locations.

2.2 Funding Requests and Alternatives

This section includes a funding request, but it is worth noting that in the event that we do not receive the funding/materials requested, we will buy them out of pocket.

Item	Qty.	Unit Cost	Purpose / Justification	Total
MeshCore Repeaters	2	\$100.99	An essential device in extending range/robustness of the network. Having at least one would allow for benchmarking/security testing with a repeater. Having two would prevent a single-point of failure (in the off-chance we break one), and would allow for more degrees of freedom in any benchmarking tests. Website: https://www.seeedstudio.com/SenseCAP-Solar-Node-P1-Pro-for-Meshtastic-LoRa-p-6412.html	\$201.98
MeshCore Nodes	10	\$26.70	An essential communication device of the network. Having two enables actual communication. Having four would allow for each member in our group to operate devices independently, which would serve to demonstrate real-world usage patterns of a small community. Having 10-17 would enable an interesting benchmarking test in which we create a (3x3 or 4x4) grid of nodes, and measure performance in a more repeatable/informative way. Website: https://heltec.org/project/mesh-node-t114/ (Need 915 MHz (US) version, plastic shell preferred, GPS essential for benchmarking, Display unessential)	\$267.00

3 Annotated Bibliography

Below is a list of references relevant to our proposed research project annotated with information about what those references are about and how they relate to the project.

1. **El Gemayel, C.; El Gemayel, J.; Constantin, J. HERMES: Metric-Driven Multi-Transport Routing for Civilian Messaging During Connectivity Disruption. Network 2026, 6, 64. <https://doi.org/10.3390/network6030064>**
Annotation: This paper introduces HERMES, which is a hybrid communication method utilizing a mix of HTTP/IP networking, Bluetooth Low Energy (BLE) mesh communication, and

Delay-Tolerant Networking (DTN). HERMES is motivated by providing those in need the ability to communicate, the motivation of this paper has many overlaps with the motivation of our own.

2. **R. Bruno, M. Conti and E. Gregori, "Mesh networks: commodity multihop ad hoc networks," in IEEE Communications Magazine, vol. 43, no. 3, pp. 123-131, March 2005, doi: 10.1109/MCOM.2005.1404606.**

Annotation: An article from 2005, that has an in-depth overview of the mesh-network landscape of the time. Some key takeaways that directly relate to our project are as follows: using mesh as an accessible means to extend internet access is not unprecedented; ad-hoc networks have a network-diameter bottleneck; all layers of an ad-hoc network must be self-manageable, self-configurable, and self-healing; fairness in ad-hoc networks is complex.

3. **J. Jubin and J. D. Tornow, "The DARPA packet radio network protocols," in Proceedings of the IEEE, vol. 75, no. 1, pp. 21-32, Jan. 1987, doi: 10.1109/PROC.1987.13702.**

Annotation: This paper is one of the first to explore a wireless method of communication via a packet switched, store-and-forward radio. Modern ad hoc networks are based on this network design for multihop message transmission. This describes how the multi-hop algorithm traverses the network topology with a table of its neighbors.

4. **S. T. Ahmed, A. A. Ahmed, A. Annamalai, and M. F. Chouikha, "A Scalable and Energy-Efficient LoRaWAN-Based Geofencing System for Remote Monitoring of Vulnerable Communities," IEEE Access, vol. 12, pp. 48540–48554, 2024, doi: 10.1109/ACCESS.2024.3383778.**

Annotation: The paper describes the implementation of a LoRaWAN geofencing system using nodes, similar in capabilities to those we propose to use. The system is also similarly designed, using central server node(s) to receive messages from mobile client nodes and passing those messages to a dedicated program for features (in the paper, alerts when a node travels outside a defined spatial perimeter). The paper describes improvements in the established ECHO protocol for LoRaWAN resulting in reduced collisions, improved battery life, increase in successful packet delivery rates - at the cost of latency. The considerations made for these improvements are relevant to design choices we will make in designing our application's protocols.

4 Team Roles

1. Natalie: privacy and security lead.
2. Carly: hardware and radio technology lead.
3. Quinn: community and social considerations lead.
4. Joel: benchmarking and testing lead.
5. All members: responsible for collaboration, design, development, and editing of:
 - Code
 - Presentation
 - Report
 - Facilitate and cooperate in conducting tests and hardware setup

5 Anticipated Timeline

What follows is a loose plan for how our group expects to spend time between Week 5 to Week 15. Note the tasks are associated with a week number.

5.1 Carly's Anticipated Timeline (Radio Hardware Focused)

A timeline of what Carly expects to focus on in Week 5 to Week 9.

5. Understand radio/mesh and its hardware
6. Understand FCC Regulations
7. Learn radio/mesh disadvantages (jamming, spoofing, etc)
8. Research radio methods to potentially overcome disadvantages
9. ... 15. Group App Development and Report/Presentation

5.2 Joel's Anticipated Timeline (Benchmarking Focused)

A timeline of what Joel expects to focus on in Week 5 to Week 9.

5. Literature Review + Hardware Setup
6. Literature Review + Design of Benchmarking Tasks
7. Benchmarking
8. Benchmarking
9. ... 15. Group App Development and Report/Presentation

5.3 Natalie's Anticipated Timeline (Security Focused)

5. ... 8. Individual Special Topic Research
9. ... 15. Group App Development and Report/Presentation

5.4 Quinn's Anticipated Timeline (Community Focused)

A timeline of what Quinn expects to focus on in Week 5 to Week 9.

5. Review MeshCore capabilities, interfacing with devices
6. Review community implementations of MeshCore
7. Review alternatives to MeshCore, tradeoffs
8. Review "internet sharing"
9. ... 15. Contribute to group deliverables

5.5 Collaborative Deliverable Development Timeline

9. System design, specifications. Begin drafting presentation/report
10. ... 13. App development and presentation/report writing
14. Final revisions of presentation, report, app
15. Finalize and deliver demo, presentation, and report

6 AI Disclosure

Our team did not use AI tools in completing this assignment. Reading references and annotations was done manually to facilitate our learning. We leave open the possibility of using AI as a feature in the production or implementation of our project, but will cite it as appropriate and with consideration to stakeholder values.